Teaching Philosophy

Alain Maubert Crotte

My Concept of Learning

I believe that students learn most effectively through active construction, not passive reception. In technology and design, learning happens when students make something that behaves, encounter the consequences of their decisions, and iterate. My doctoral research on left-handed inclusion in computer games taught me that learning requires recognizing the invisible defaults in our tools; students learn deeply when they are asked to critically examine who technology serves and how interfaces include or exclude.

My Approach to Teaching

My approach is collaborative, adaptive, and grounded in interdisciplinary practice. Having taught across the Faculty of Science, the Faculty of Media, Art, and Performance, and the Faculty of Arts, I bridge different disciplinary languages. I act as a facilitator rather than a lecturer, guiding students through the problem-solving process in real time. I bring applied industry and research experience from my time as Head of Mixed Reality at the ReACT Lab directly into the classroom.

Definition of Good Teaching

Good teaching demystifies complex processes and creates equitable access to learning. It does not simply transfer information; it builds confidence and technical auton-

omy. A good teacher ensures that no student is left behind due to a lack of access to hardware or prior technical background, and creates an environment where failure is treated as a necessary step in the design cycle.

Teaching and Learning Goals

- **Technical Autonomy:** I want students to leave my courses with the confidence to troubleshoot, learn new software independently, and apply technical skills to creative problems.
- **Critical Awareness:** In game and interactive design courses (such as CTCH / DES 312), I aim for students to examine representation, user agency, and accessibility alongside technical work, treating these as design requirements rather than afterthoughts.
- **Interdisciplinary Competence:** I prepare students to navigate both the technical and creative sides of projects, bridging the gap between computer science implementation and media arts practice.

Teaching Methods and Strategies

- **Live Coding and Demonstrations:** I demonstrate concepts live. *(Example: In CTCH 223 / CS 207, CS 215, and CTCH / DES 312, students see the coding process in real time and ask questions as issues arise).*
- **Equitable Teamwork:** When teaching with specialized hardware, students occasionally face technical hurdles or missing components. I form balanced teams where each student has a specific role, such as operating equipment, developing assets, or documenting processes, and I encourage peer support to share

missing parts. *(Example: In CTCH / DES 312, this ensures learning continues uninterrupted even if a student's equipment fails or is incomplete).*

- **Experiential Learning:** I move the classroom into the community. *(Example: In MAP 102, I adapted the curriculum to require site visits to cultural institutions across Regina, allowing students to critically interact with the city's creative organizations).*
- **Applied Research Integration:** I bring my applied research prototypes into the classroom as active case studies. By examining my design of the VisionAR: Impairment Simulator and the VR Regalia: Museum Interaction Prototype, students learn how to translate theoretical concepts of accessibility, spatial interaction, and haptic feedback into functioning, user-centered interfaces.

My Belief of Assessments

Assessment should measure the process of learning, not just the final polish. I advocate for flexible assessment methods beyond traditional exams, using group projects, iterative rubrics, and process documentation requirements. Students are evaluated on their ability to articulate why a design or technical decision was made. *(Example: When students in CTCH / DES 312 struggled with the final project, I provided supplementary Unity documentation and extended deadlines when necessary to ensure comprehension over strict pacing).*

Self-Reflection

I actively review student feedback to improve my delivery. Early in my teaching career (CS 100, CS 215), feedback indicated that the dense volume of material sometimes felt challenging to absorb within a single class period. I have since heavily revised my

approach, shifting from dense lectures toward interactive problem-solving and live coding. As a lifelong learner, I continuously experiment with assessment structures (such as moving from high-stakes exams to iterative project rubrics) to better measure practical skill acquisition.

- View my full summary of teaching evaluations (PDF)

Whether teaching foundational computer science or advanced creative technologies, I want students to see themselves not just as users of technology, but as shapers of a more inclusive, creative, and compassionate digital future.